☑ CTL Provisional DECISION SHEET (PDSH)
☐ CTL DECISION SHEET (DSH)

Standard(s) (incl. year)	**Subclause(s)**	**Tracking No.**
IEC 60947-1:2024 Ed.6	9.3.4.1.2 b) and Fig. 9 – 11 Test circuits	2287
Category		
POW		
Subject	**Keywords**	**Developed by**
Test circuit – use of shunt resistors for air-cored reactors	Shunt resistor, test circuit, air-cored reactors	ETF 7
Question		
Can shunt resistances be omitted from air-core reactors when they are located on the high voltage side of the supply circuit?		
Decision		
Shunt resistors can be omitted when X is located on the high-voltage side of the supply circuit.		
Explanatory notes		
Enquiry: Date:	2025-04-24	
PDSH: Year of publication: To be approved at CTL plenary:	2025 2026	
DSH Publication date:	---	
Transient recovery voltages (TRV) occur when a high current test circuit is interrupted and depend on the characteristics of the test circuit and the interrupting device. In order to damp the TRV IEC 60947-1: 2024 requires to shunt air-core resistors (0.6% of current). This has an effect when X is located on the low voltage side of the test circuit supply. It does not have an effect when X is located on the high-voltage side of the test circuit supply, due to the transformer transmission behavior.		

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